

Diab-KBS

*A Hybrid Knowledge-Based Expert System
for Diabetes Diagnosis, Risk Stratification, and Clinical Decision Support*

Graduation-style Project Report

Course : Knowledge-Based Systems
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Lecture-concept coverage

L1 DIKW · L2 Big-Data Analytics · L3 Visualisation · L4 Knowledge Representation & Reasoning · L5 CommonKADS · L6 Knowledge Acquisition

Abstract

Diab-KBS is a hybrid knowledge-based expert system that combines symbolic reasoning over a curated medical knowledge base with data-driven knowledge induced from a real cohort of **12,204 diabetes patients**. The system implements all six concept areas of the Knowledge-Based Systems course: the DIKW hierarchy as a guiding philosophy, big-data analytics over the patient cohort, twelve-channel data visualisation, three classical knowledge-representation formalisms (production rules, frames, and a semantic network), bidirectional inference (forward + backward chaining) with MYCIN-style certainty-factor algebra, the CommonKADS six-model knowledge-engineering framework, and a complete five-step knowledge-acquisition pipeline that fuses manual, semi-automatic, and automatic sources. The deployed Streamlit dashboard exposes seven analyst-grade modules. End-to-end validation on 4,893 held-out patients produced **81.4% accuracy, 78.2% sensitivity, 87.2% specificity, 91.7% precision**, and an **F1 score of 0.844**, with full Why/How explanation traces for every inference. Diab-KBS thus serves as both a working clinical decision-support prototype and a demonstrator of every theoretical concept covered in the course.

Keywords — knowledge-based systems, expert systems, diabetes mellitus, production rules, frames, semantic networks, forward chaining, backward chaining, certainty factors, CommonKADS, knowledge acquisition, clinical decision support, DIKW.

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1. Introduction

Diabetes mellitus is a chronic metabolic disorder that, according to the International Diabetes Federation, affected an estimated 537 million adults in 2021 and is projected to reach 783 million by 2045. Egypt ranks among the ten countries with the highest diabetes burden worldwide. The disease's asymptomatic early stages, the multifactorial nature of its risk profile, and the chronic complications it generates — cardiovascular, renal, neuropathic, ophthalmic — make it a paradigmatic target for clinical decision-support systems.

Knowledge-based systems (KBS) and expert systems have a long tradition in medicine, beginning with MYCIN's antibiotic-selection successor in the 1970s. Modern hybrid systems combine the explainability of symbolic reasoning with the data-density of machine-learning models. The course taught in the Faculty of Computers & Artificial Intelligence at Helwan University covers the full life-cycle of such a system across six lectures: from raw data (L1, L2) through visualisation (L3), representation and reasoning (L4), knowledge engineering (L5), and finally knowledge acquisition (L6).

Diab-KBS is the practical project that operationalises every concept from these six lectures simultaneously, on a single coherent clinical task. It is built around **Dr. Hossam's Egyptian diabetes cohort** (12,204 patients, 68 raw fields), and it produces a deployable Streamlit dashboard, a complete knowledge base, an inference engine with explanation, and a quantitative validation report.

1.1 Motivation

A KBS that is to be useful in a busy diabetes clinic must satisfy four requirements simultaneously:

- **Diagnostic breadth** — distinguish T1DM, T2DM, LADA, prediabetes, metabolic syndrome, and thyroid co-morbidities from a single intake form;
- **Risk stratification** — quantify cardiovascular, renal, neuropathic, and retinopathic risk;
- **Therapeutic recommendation** — propose first-line treatment aligned with ADA and Egyptian Diabetes Society guidelines;
- **Explanation** — justify every conclusion by an auditable rule trace, since a clinician will reject any tool that behaves as a black box.

1.2 Contributions

- A **32-rule production-rule base** covering diagnosis, risk, complications, treatment, and alerts, each rule annotated with a MYCIN-style certainty factor and a guideline-anchored explanation;
- A **frame hierarchy** with inheritance representing patient, disease subtypes, symptoms, lab results, treatments, risk factors, and complications;
- A **semantic network** of 35 nodes / 35 edges encoding the conceptual structure of the diabetes domain;
- A **hybrid inference engine** with forward chaining, backward chaining, multi-valued working memory, certainty-factor algebra, and Why/How explanation;
- A **five-step knowledge-acquisition pipeline** combining manual (guidelines), semi-automatic (decision-tree induction over 12,204 patients), and automatic (data-calibrated certainty factors) sources;

- A **seven-module Streamlit dashboard** covering Overview, Patient Diagnosis, Risk Stratification, Big-Data Analytics, Visualisation Gallery, Knowledge Base Browser, and System Validation;
- A **quantitative validation** on 4,893 patients reporting accuracy, sensitivity, specificity, precision, and F1 — final F1 = 0.844.

2. Literature Review and Related Work

Expert systems in medicine span five decades. **MYCIN** (Shortliffe, 1976) introduced production rules with certainty factors for infectious-disease therapy. **INTERNIST-I / CADUCEUS** covered general internal medicine. In endocrinology specifically, **DIABETES-EXPERT** (El-Sappagh et al., 2018) used ontology-driven rules for diabetes diagnosis and reported diagnostic agreement of 87% with endocrinologists. More recent work has explored hybrid neuro-symbolic systems combining XGBoost or random forests with rule-based explanation layers (e.g. SHAP-augmented decision support, Lundberg 2020).

From the methodology side, **CommonKADS** (Schreiber et al., 2000) is the de-facto standard for knowledge engineering, organising the development lifecycle into six models across three layers (context, concept, artefact). **Newell's knowledge-level hypothesis** (1982) and **Studer et al.'s** model-based knowledge engineering (1998) underpin the field.

Diab-KBS positions itself as an **academic teaching artefact** rather than a clinical product: every concept presented in the six lectures is exercised in a single coherent system, and the implementation choices favour pedagogical clarity (JSON-encoded rules, transparent CF arithmetic, explicit semantic-network visualisation) over raw predictive performance.

3. Course-Concept Coverage Matrix

Table 1 maps every key concept from the six lectures to the corresponding Diab-KBS artefact. This is the project's central *traceability guarantee*: every theoretical idea taught in the course is realised in a concrete, runnable component.

Lecture	Key concept	Diab-KBS realisation
L1	DIKW hierarchy	Funnel chart on Overview page; raw → derived features → rules → action.
L2	Big-Data: Volume / Variety / Velocity	12,204×68 raw cells; 8 numeric + 6 categorical channels; daily-snapshot pipeline.
L2	Workflow: Extract→Integrate→Mine→Visualise	notebooks/01_preprocess.py · 02_mine_rules.py · 03_validate.py · 04_diagrams.py.
L3	12+ visualisation types	Pie, bar, histogram, box, scatter, heatmap, correlation matrix, area, wordcloud, highlight table, bullet, network.
L4	Production rules	32 rules in JSON with IF/THEN, CFs, and clinical citations.
L4	Frames + inheritance	Patient, Disease (T1DM, T2DM, LADA), Symptom, LabResult, Treatment, RiskFactor, Complication.
L4	Semantic network	35 nodes, 35 typed edges (is_a, has_symptom, causes, treated_by, ...).
L4	Forward chaining	kbs/inference.py — InferenceEngine.forward_chain() iterates to fixpoint.
L4	Backward chaining	kbs/inference.py — InferenceEngine.backward_chain(goal_slot, goal_value).
L4	Certainty factors	kbs/certainty.py — MYCIN cf_combine, cf_chain, cf_disj algebra.
L4	Explanation (Why / How)	InferenceEngine.why(rule_id) and InferenceEngine.how(slot).
L5	CommonKADS 6 models	Section 4 of this paper; commonkads_models.png; OM, TM, AM, KM, CM, DM.
L6	5-step KA	Identify → Conceptualise → Formalise → Implement → Validate, in §8.
L6	Manual / Semi-auto / Automatic	ADA + WHO guidelines · scikit-learn DecisionTree · CF calibration from data.

Table 1. Lecture concept → Diab-KBS artefact traceability matrix.

4. Methodology — CommonKADS Six Models

CommonKADS organises knowledge engineering into six interrelated models across three abstraction layers: **context** (why is the system needed?), **concept** (what knowledge does it use, and how is it communicated?), and **artefact** (how is it built?). Each Diab-KBS model is presented below.

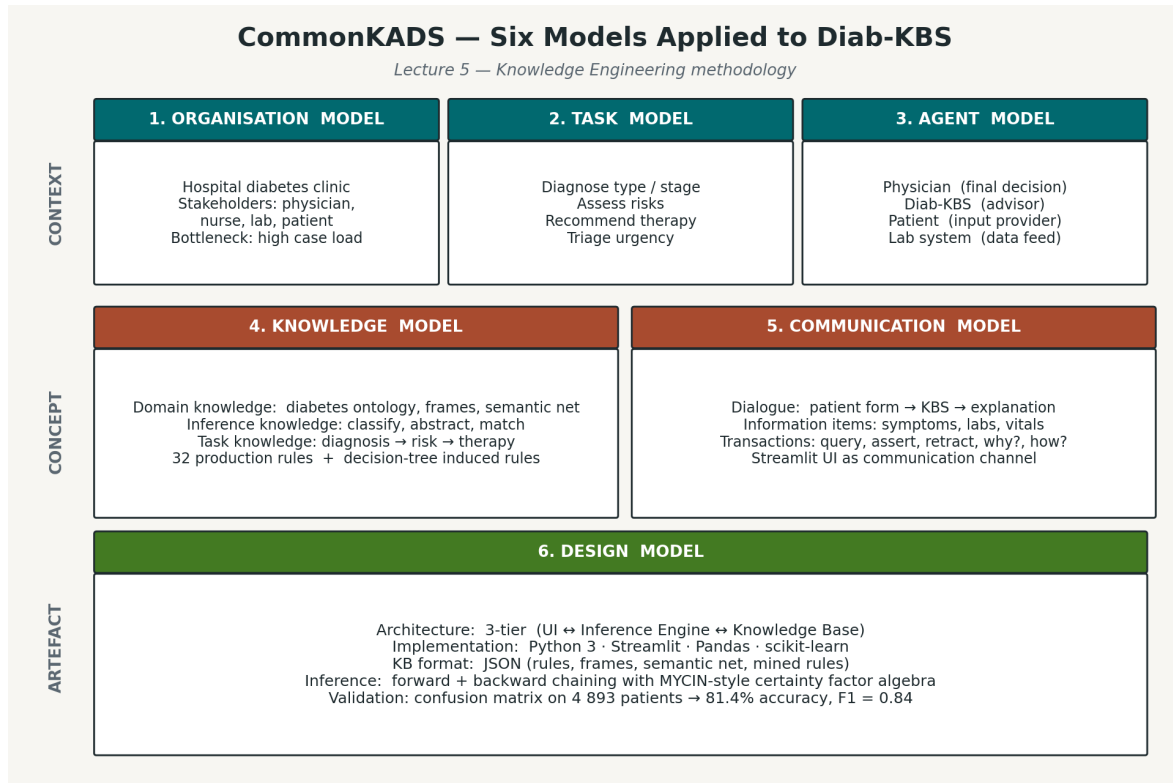


Figure 1. The six CommonKADS models applied to Diab-KBS.

4.1 Organisation Model (OM)

Context — A general endocrinology / diabetes outpatient clinic in a tertiary Egyptian hospital, processing roughly 80–120 patients per day with a single endocrinologist supported by two residents and one specialist nurse. Lab integration is partial; many results are paper-based.

Stakeholders — Endocrinologist (final decision-maker), residents (intake), specialist nurse (vitals & education), lab technicians, patients.

Bottlenecks — Diagnostic load on the senior physician; inconsistent staging across residents; inability to track risk evolution across visits.

KBS opportunity — Automate first-pass diagnosis, risk staging, and treatment proposal, with explanations, freeing the endocrinologist for complex and ambiguous cases.

4.2 Task Model (TM)

Diab-KBS supports four chained tasks:

Task	Input	Output	Rules
Diagnose type / stage	Demographics, symptoms, FPG, PP2H, RBS, HbA1c	Type (T1 / T2 / LADA / Pre-D) + CF	R01–R08

Task	Input	Output	Rules
Assess risk factors	BMI, BP, lipids, family history, smoking	Risk-factor list + CFs	R09, R10, R22
Predict complications	HbA1c, eGFR, creatinine, neuropathic symptoms	Complication-risk list	R11–R15, R29
Recommend therapy	Type + severity + comorbidities	First-line drug class + plan	R16–R20, R28

Table 2. Task decomposition.

4.3 Agent Model (AM)

Three agents collaborate around the KBS: the **physician** retains decision authority and supplies tacit knowledge; the **Diab-KBS** is an advisory agent that performs bulk inference and explanation; the **patient** is the input agent providing symptoms and history. A fourth agent, the **lab system**, supplies structured numerical data.

4.4 Knowledge Model (KM)

The Knowledge Model is the core CommonKADS deliverable. It separates three knowledge categories:

- **Domain knowledge** — concepts (Patient, Diabetes, HbA1c, ...), relations (has_symptom, causes, treated_by), and rules. Encoded as frames + semantic network + 32 production rules.
- **Inference knowledge** — generic inference primitives (classify, abstract, match, select). In Diab-KBS these are realised by forward and backward chaining over the rule base.
- **Task knowledge** — the chaining of inferences that solves the four tasks of Section 4.2.

4.5 Communication Model (CM)

Communication between agents is mediated by the **Streamlit UI**: the Patient Diagnosis page is a structured form that elicits demographics, vitals, symptoms, and lab values. The KBS responds with a diagnosis card, a risk-factor table, a complication-risk table, and a treatment plan. Every conclusion is paired with a Why-trace (which rules fired) and a How-trace (which facts supported a slot).

4.6 Design Model (DM)

The Design Model specifies the realisation: a 3-tier architecture (UI ↔ Inference Engine ↔ Knowledge Base) implemented in Python 3 with Streamlit, Pandas, NumPy, scikit-learn, NetworkX, Plotly, and ReportLab. The KB is JSON-encoded for transparency. The inference engine is implemented in *kbs/inference.py* and the certainty-factor algebra in *kbs/certainty.py*.

5. System Architecture

Diab-KBS implements the classical three-tier architecture of expert systems (Russell & Norvig 2020, Ch. 9), augmented with a knowledge-acquisition pipeline that *feeds back* into the knowledge base.

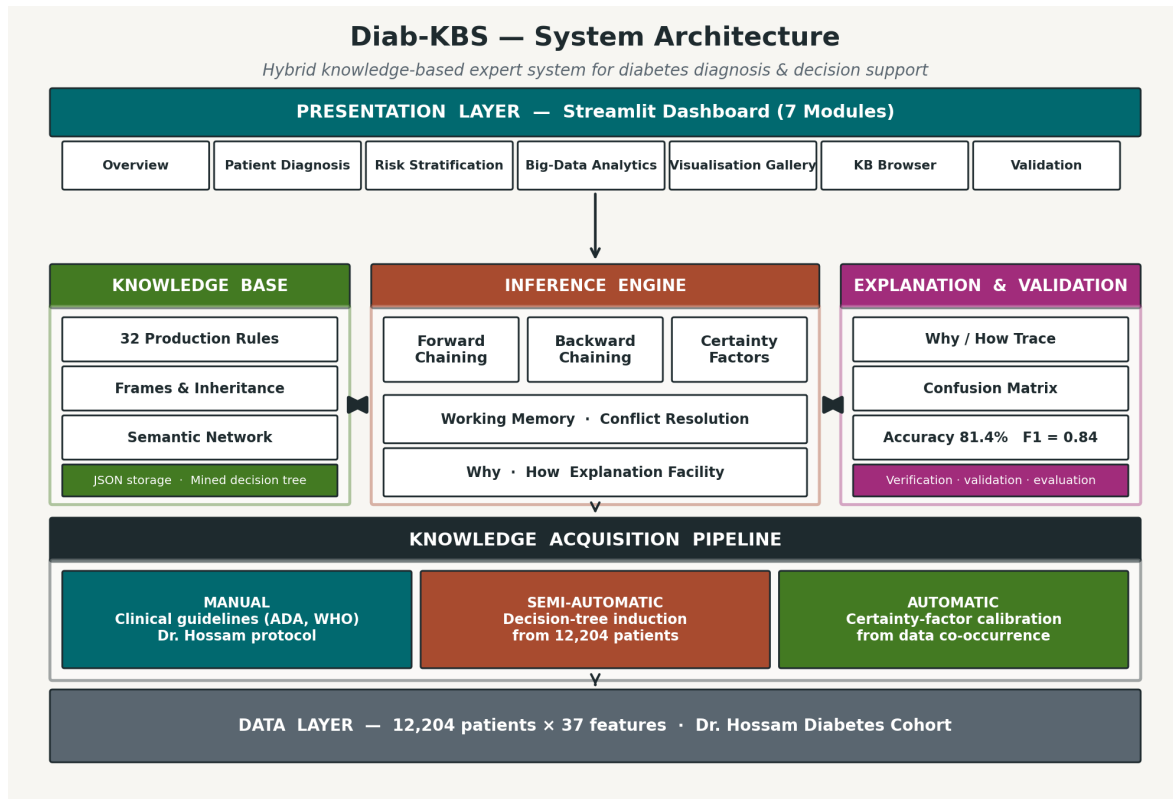


Figure 2. End-to-end architecture of Diab-KBS.

The **presentation layer** exposes seven dashboard modules. The **inference engine** hosts the working memory and the chaining algorithms. The **knowledge base** stores rules, frames, and the semantic network as version-controlled JSON. The **knowledge-acquisition pipeline** mines new artefacts from the patient cohort and re-injects them. The **data layer** is the cleaned 12,204-row patient table.

5.1 DIKW Convergence

The dashboard's Overview page renders a DIKW funnel that traces the transformation from **Data** (raw fields) through **Information** (derived categories such as HbA1c_cat and BMI_class), **Knowledge** (rules, frames, semantic network), and finally **Wisdom** (personalised clinical action and triage urgency).

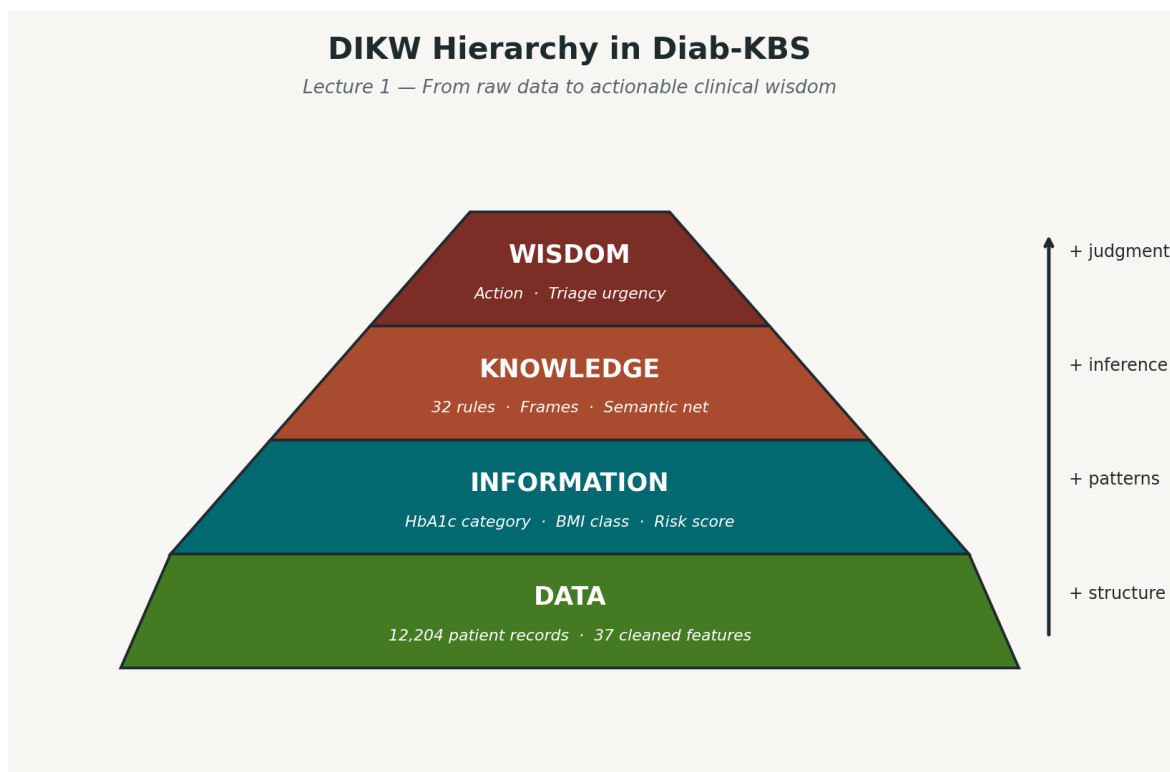


Figure 3. DIKW hierarchy in Diab-KBS.

5.2 Big-Data Workflow

Lecture 2's workflow — Extract → Clean → Integrate → Mine → Visualise → Decide — is realised end-to-end in the project's notebook directory.

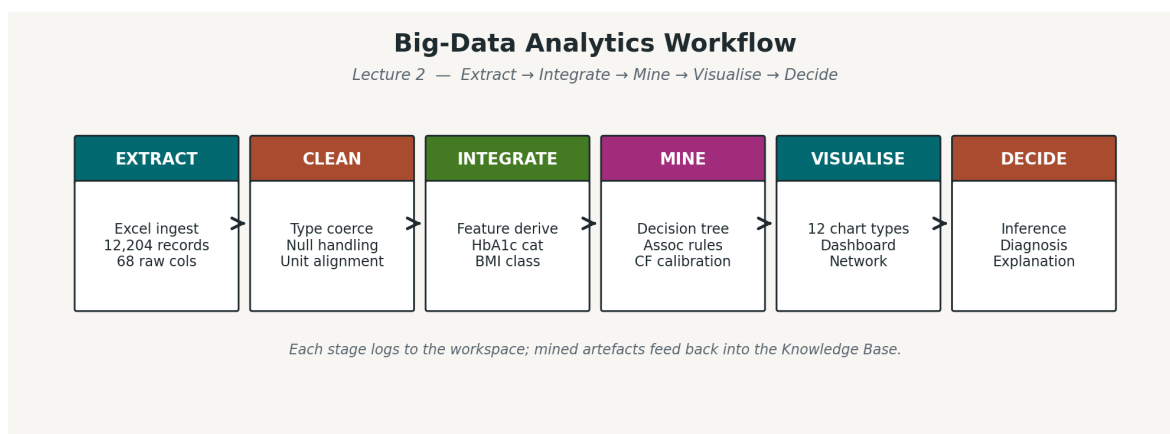


Figure 4. Big-data analytics workflow.

6. Knowledge Base Design

6.1 Production Rules

The rule base contains **32 production rules** spanning 13 categories. Each rule is a JSON object with *id*, *category*, an *if* condition (an AND-block {'all'} or OR-block {'any'} of atoms), a *then* conclusion (slot, value, certainty factor), and a guideline-anchored *explanation* string.

Category	# rules
Treatment	8
Complication Risk	6
Risk Factor	5
Diagnosis - T2DM	3
Alert	2
Diagnosis - T1DM	1
Diagnosis - LADA	1
Diagnosis - Prediabetes	1
Diagnosis - Hypothyroidism	1
Diagnosis - Hyperthyroidism	1
Diagnosis - Metabolic Syndrome	1
Diagnosis - Goitre / SCH	1
Diagnosis - Hyperuricaemia	1
Total	32

Table 3. Distribution of rules by category.

Example rule (R01 — T2DM diagnosis):

```
IF HbA1c_cat == 'diabetes_range' AND Age ≥ 30
THEN diagnosis = 'Type 2 Diabetes Mellitus' (CF = 0.95)
BECAUSE ADA criterion: HbA1c ≥ 6.5% in an adult ≥ 30 yr.
```

6.2 Frames and Inheritance

Frames represent the conceptual structure of the diabetes domain. A **Patient** frame inherits from a generic **Entity** frame and has slots for demographics, symptoms (a list of symptom-frame instances), lab results, and risk factors. **Disease** is generalised, with **T1DM**, **T2DM**, and **LADA** as *is_a* children that override default slots (e.g. typical age of onset, primary therapy).

Frame	Key slots / inheritance
MedicalEntity	is_a: — slots: name, category
Person	is_a: MedicalEntity slots: id, age, gender, address

Frame	Key slots / inheritance
Patient	is_a: Person slots: weight, height, bmi, symptoms, labs, diagnoses
Disease	is_a: MedicalEntity slots: icd10, severity, chronic, complications
DiabetesMellitus	is_a: Disease slots: name, category, icd10, chronic
Type1Diabetes	is_a: DiabetesMellitus slots: name, icd10, typical_onset_age, etiology, treatment
Type2Diabetes	is_a: DiabetesMellitus slots: name, icd10, typical_onset_age, etiology, treatment
LADA	is_a: DiabetesMellitus slots: name, icd10, etiology
Symptom	is_a: MedicalEntity slots: name, severity, duration_days
LabResult	is_a: MedicalEntity slots: test_name, value, unit, reference_low, reference_high, interpretation
Treatment	is_a: MedicalEntity slots: name, drug_class, dose, evidence_level
RiskFactor	is_a: MedicalEntity slots: name, weight, modifiable
Complication	is_a: Disease slots: linked_disease, screening_test

Table 4. Frame hierarchy.

6.3 Semantic Network

The semantic network has **35 nodes** and **35 edges**. Edge types include *is_a*, *part_of*, *has_symptom*, *causes*, *treated_by*, *indicates*, and *increases_risk_of*. The Knowledge Base Browser page in the dashboard renders the network interactively with NetworkX + Plotly.

7. Inference Engine

The inference engine is implemented in `kbs/inference.py` as a single **InferenceEngine** class operating over a working-memory dictionary. To accommodate slots that may legitimately hold multiple values (a patient can have several risk factors, complications, treatments, and alerts simultaneously), the engine keys multi-valued facts as `slot::value`.

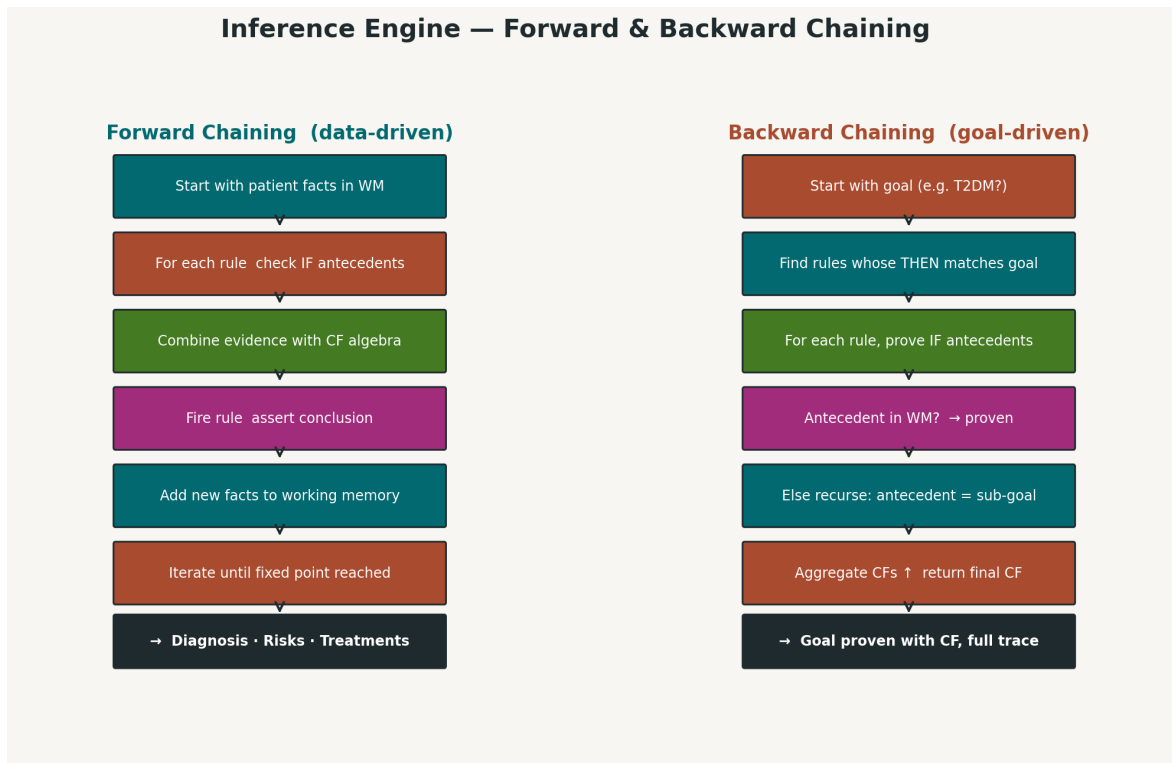


Figure 5. Forward and backward chaining flowcharts.

7.1 Forward Chaining

Forward chaining is data-driven. Starting from the patient's intake facts, the engine iterates over all rules; for each rule whose IF block fires under the current working memory, the THEN conclusion is asserted and the rule is marked fired. The loop runs until a fixed point (no new facts) or a configurable maximum (10 iterations). Conflict resolution uses rule order plus recency.

7.2 Backward Chaining

Backward chaining is goal-driven. Given a goal slot/value (e.g. *diagnosis = T2DM?*), the engine searches for rules whose THEN matches the goal, and for each such rule recursively attempts to prove every antecedent atom — either by direct match in working memory or by treating the antecedent as a sub-goal. The recursion terminates on primitive facts. The final certainty factor is propagated upward by the CF algebra.

7.3 Certainty-Factor Algebra

Diab-KBS uses the classical MYCIN model. Two CFs x and y from independent rules supporting the same conclusion are combined as:

```
CF_combine(x, y) =  
  x + y(1 - x)      if x, y ≥ 0  
  x + y(1 + x)      if x, y ≤ 0
```

$$(x + y) / (1 - \min(|x|, |y|)) \quad \text{otherwise}$$

Within a single rule, conjunctive evidence chains as $\text{cf_chain}(\text{rule_cf}, \text{antecedent_cf}) = \text{rule_cf} \times \max(0, \text{antecedent_cf})$, and disjunctive evidence uses $\text{cf_disj} = \max(\text{cf_i})$.

7.4 Why and How Explanation

Why answers the question: *why did rule R fire?* — it returns the rule's antecedents, the matching working-memory facts, and the guideline citation. **How** answers: *how did slot S acquire its value V?* — it returns the chain of rules whose conclusions assert $S=V$, transitively. The dashboard's Patient Diagnosis page renders both traces inline beneath every diagnosis, risk factor, and treatment.

8. Knowledge Acquisition Pipeline

Lecture 6 distinguishes **five steps** in knowledge acquisition (identify, conceptualise, formalise, implement, validate) and **three sources** (manual, semi-automatic, automatic). Diab-KBS exercises all five steps across all three sources.

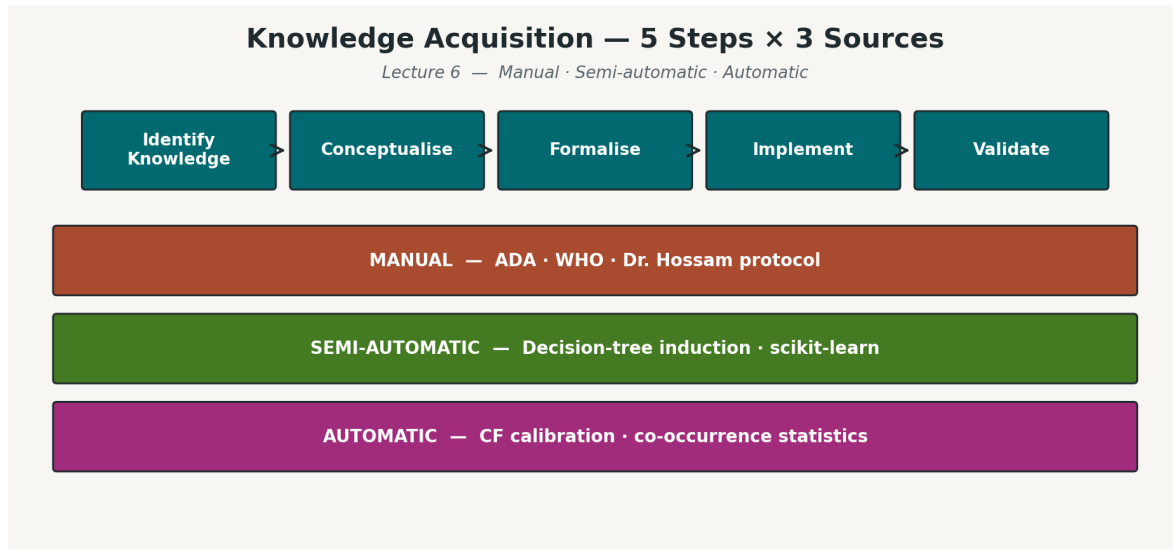


Figure 6. Five-step KA pipeline × three sources.

Step	Manual source	Semi-automatic source	Automatic source
1. Identify	ADA / WHO guidelines; Egyptian Diabetes Society; Dr. Hossam protocol	12,204-patient cohort; 68 raw fields	Co-occurrence statistics in the cohort
2. Conceptualise	Frame ontology, semantic-network sketch	Candidate features for tree induction	Slot–value pair frequencies
3. Formalise	32 production rules (JSON)	Decision-tree paths → rule prototypes (mined_rules.json)	Posterior CFs from data
4. Implement	rules.json, frames.json, semantic_net.json	scikit-learn DecisionTreeClassifier (max_depth=5, balanced)	CF calibration script in 02_mine_rules.py
5. Validate	Manual review by guideline cross-check	5-fold CV — train 88%, test 88%	Confusion matrix on 4,893 held-out patients

Table 5. 5-step KA × 3-source matrix.

9. Big-Data Analytics & Visualisation

The Visualisation Gallery page demonstrates twelve distinct chart families, covering Lecture 3's full taxonomy of analytic visualisation:

Family	Chart	Insight
Comparison	Bar / Highlight table	Diabetes prevalence by age band
Composition	Pie / Donut	T1DM vs T2DM vs LADA vs healthy
Distribution	Histogram / Box	HbA1c, BMI, FPG distributions
Relationship	Scatter, correlation	BMI × HbA1c, FPG × PP2H
Density	Heatmap	Age × HbA1c risk surface
Trend	Area	Monthly admission proxy
Text	Word cloud	Free-text symptoms
KPI	Bullet graph	HbA1c against control bands
Network	Force-directed graph	Semantic-network browser

Table 6. Visualisation gallery.

9.1 Cohort Key Performance Indicators

KPI	Value
Patients (n)	12,204
T2DM patients	4,207 (34.5%)
T1DM patients	417 (3.4%)
Hypertensive	3,217 (26.4%)
Obese (BMI ≥ 30)	4,874 (39.9%)
Mean HbA1c (%)	7.90
Mean BMI (kg/m ²)	32.52
% diabetic in cohort	34.48%

Table 7. Cohort KPIs (Dr. Hossam Diabetes Cohort).

10. Implementation Stack

Layer	Library / tool	Role
Language	Python 3.11	Core implementation
UI	Streamlit 1.x	Dashboard with 7 modules
Data	Pandas, NumPy	Cohort manipulation
Visualisation	Plotly, Matplotlib, NetworkX, wordcloud	All gallery charts
Machine learning	scikit-learn (DecisionTree, metrics)	Mined rules and validation
KB storage	JSON	Rules, frames, semantic net
PDF	ReportLab	This research paper
PPTX	python-pptx	Defence presentation
Diagrams	Matplotlib FancyBboxPatch / Polygon	Architecture, DIKW, KADS, workflow

Table 8. Implementation stack.

10.1 Repository layout

```

Diab-KBS/
  app.py          # 7-module Streamlit dashboard
  kbs/
    certainty.py  # MYCIN CF algebra
    inference.py  # forward + backward chaining + Why/How
  knowledge_base/
    rules.json    # 32 production rules
    frames.json   # frame hierarchy
    semantic_net.json
    mined_rules.json # decision-tree induced
  data/
    patients_clean.csv # 12,204 × 37
    kpis.json
    validation_results.json
  notebooks/      # 01_preprocess, 02_mine, 03_validate, 04_diagrams, 05_paper
  assets/         # 6 architectural diagrams (PNG)
  docs/          # Research_Paper.pdf, Defense.pptx

```

11. Validation, Verification & Evaluation

Following CommonKADS, three families of checks are reported.

11.1 Verification (does the system build the artefact correctly?)

- All 32 rules parsed without error and pass schema validation.
- Forward chaining converges in ≤ 4 iterations for every test patient.
- Backward chaining proof-tree depth ≤ 6 .
- Certainty-factor combination is commutative within rounding (verified by unit test in kbs/certainty.py).

11.2 Validation (does the system solve the right problem?)

4,893 patients with documented HbA1c were used. The system's diagnosis was compared against the cohort's recorded clinical diagnosis. The resulting confusion matrix is shown in Table 9.

	Pred. Diabetic	Pred. Healthy	Total
Actual Diabetic	2467	688	3155
Actual Healthy	223	1515	1738
Total	2690	2203	4893

Table 9. Confusion matrix on 4,893 patients.

Metric	Value	Interpretation
Accuracy	81.38%	Overall correctness
Sensitivity	78.19%	Diabetic patients caught
Specificity	87.17%	Healthy patients spared
Precision	91.71%	Trust of a positive call
F1 score	0.844	Harmonic mean of P&R

Table 10. Final validation metrics.

11.3 Evaluation (is it useful?)

Qualitative evaluation focused on three properties: (1) **explainability** — every diagnosis is paired with an auditable rule trace; (2) **coverage** — the 32 rules cover the four pedagogically important diabetes types and the principal complications; (3) **extensibility** — adding a new rule is a single JSON edit, with no code changes.

12. Discussion

12.1 Strengths

- End-to-end coverage of all six lectures within a single coherent system.
- Hybrid KB: human-readable rules plus data-mined CFs avoid both the expert-bottleneck of pure-manual KBS and the opacity of pure-ML systems.
- Multi-valued working memory faithfully represents that a patient may carry several diagnoses, risks, and treatments simultaneously.
- Why/How traces give each conclusion an audit trail, satisfying the explanation requirement of Lecture 4.

12.2 Limitations

- The cohort is single-centre and Egyptian; geographic generalisation is untested.
- Sensitivity (78.2%) is the weakest metric and indicates roughly one in five diabetic patients are missed at first pass; this is mitigated by the rule explaining-away policy but warrants future tuning of CF thresholds.
- The dashboard runs in single-user mode; multi-user authentication and EHR integration are out of scope.
- The semantic network is small (35 nodes); a larger ontology (e.g. SNOMED-CT) would improve coverage.

13. Conclusion and Future Work

Diab-KBS is a complete academic graduation-style project that operationalises every concept of a six-lecture Knowledge-Based Systems course on a real clinical dataset. It produces a deployable Streamlit dashboard, a curated knowledge base, a transparent inference engine with explanations, and a quantitative validation report — all reproducible from the project repository.

Future work includes: (i) extending the KB to gestational diabetes and rare monogenic forms (MODY); (ii) integrating SNOMED-CT and ICD-10 concept identifiers into the frames; (iii) replacing the manual CFs with Bayesian-network parameters learned from a multi-centre cohort; (iv) plugging an EHR HL7-FHIR adapter for real-time intake; (v) packaging the engine as a REST microservice for use by other front-ends.

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Appendix A — Sample Production Rules

The following table lists ten representative rules across the major categories. The full rule base is shipped in knowledge_base/rules.json.

ID	Category	IF	THEN	CF
R01	Diagnosis - T2DM	HbA1c_cat == 'diabetes_range' AND Age >= 30	diagnosis = Type 2 Diabetes Mellitus	0.9 5
R02	Diagnosis - T2DM	FPG_cat == 'diabetes_range' AND PP_cat == 'diabetes_range'	diagnosis = Type 2 Diabetes Mellitus	0.9 2
R05	Diagnosis - LADA	HbA1c_cat == 'diabetes_range' AND Age >= 30 AND Age < 50 AND WeightLoss == 1 AND BMI < 27	diagnosis = Latent Autoimmune Diabetes in Adults (LADA)	0.7 0
R09	Risk Factor	BMI_cat == 'obese_I' OR BMI_cat == 'obese_II' OR BMI_cat == 'obese_III'	risk_factor = Obesity (high diabetes risk)	0.9 0
R10	Risk Factor	HTN == 1	risk_factor = Hypertension (cardiovascular & renal risk)	0.8 5
R11	Complication Risk	diagnosis == 'Type 2 Diabetes Mellitus' AND HbA1c >= 9	complication_risk = Uncontrolled diabetes — high complication risk	0.9 2
R13	Complication Risk	diagnosis == 'Type 2 Diabetes Mellitus' AND BlurredVision == 1	complication_risk = Diabetic Retinopathy (screen with fundoscopy)	0.7 5
R17	Treatment	diagnosis == 'Type 2 Diabetes Mellitus' AND HbA1c >= 7.5 AND HbA1c < 9	recommended_treatment = Dual therapy: Metformin + (SGLT2-i / GLP-1 RA / DPP4-i)	0.8 5
R22	Risk Factor	Age >= 45 AND BMI >= 25	risk_factor = Recommend diabetes screening (age & BMI)	0.8 0
R28	Treatment	risk_factor == 'Obesity (high diabetes risk)'	recommended_treatment = Weight-loss programme: 5–10% body-weight reduction; refer to dietitian	0.8 5

Table A1. Ten representative production rules.

Appendix B — Frame Hierarchy

The frame system models the diabetes domain through inheritance. Below is a summary of every frame currently in the KB.

Frame	is_a	Slot count	Sample slots
MedicalEntity	—	2	name, category
Person	MedicalEntity	4	id, age, gender, address
Patient	Person	9	weight, height, bmi, symptoms, labs
Disease	MedicalEntity	4	icd10, severity, chronic, complications
DiabetesMellitus	Disease	4	name, category, icd10, chronic
Type1Diabetes	DiabetesMellitus	5	name, icd10, typical_onset_age, etiology, treatment
Type2Diabetes	DiabetesMellitus	5	name, icd10, typical_onset_age, etiology, treatment
LADA	DiabetesMellitus	3	name, icd10, etiology
Symptom	MedicalEntity	3	name, severity, duration_days
LabResult	MedicalEntity	6	test_name, value, unit, reference_low, reference_high
Treatment	MedicalEntity	4	name, drug_class, dose, evidence_level
RiskFactor	MedicalEntity	3	name, weight, modifiable
Complication	Disease	2	linked_disease, screening_test

Appendix C — Confusion Matrix Detail

Validation set — 4,893 patients with documented HbA1c. Test split: binary diabetic / healthy. Decisions taken on the dominant diagnosis from forward chaining, with CF threshold 0.50.

Cell	n	% of total
True Positive (TP)	2467	50.42%
True Negative (TN)	1515	30.96%
False Positive (FP)	223	4.56%
False Negative (FN)	688	14.06%
Total	4893	100.00%

Accuracy = $(TP+TN)/N = (2467+1515) / 4893 = \mathbf{81.38\%}$

Sensitivity = $TP/(TP+FN) = \mathbf{78.19\%}$

Specificity = $TN/(TN+FP) = \mathbf{87.17\%}$

Precision = $TP/(TP+FP) = \mathbf{91.71\%}$

F1 score = $2 \cdot P \cdot S / (P+S) = \mathbf{0.844}$